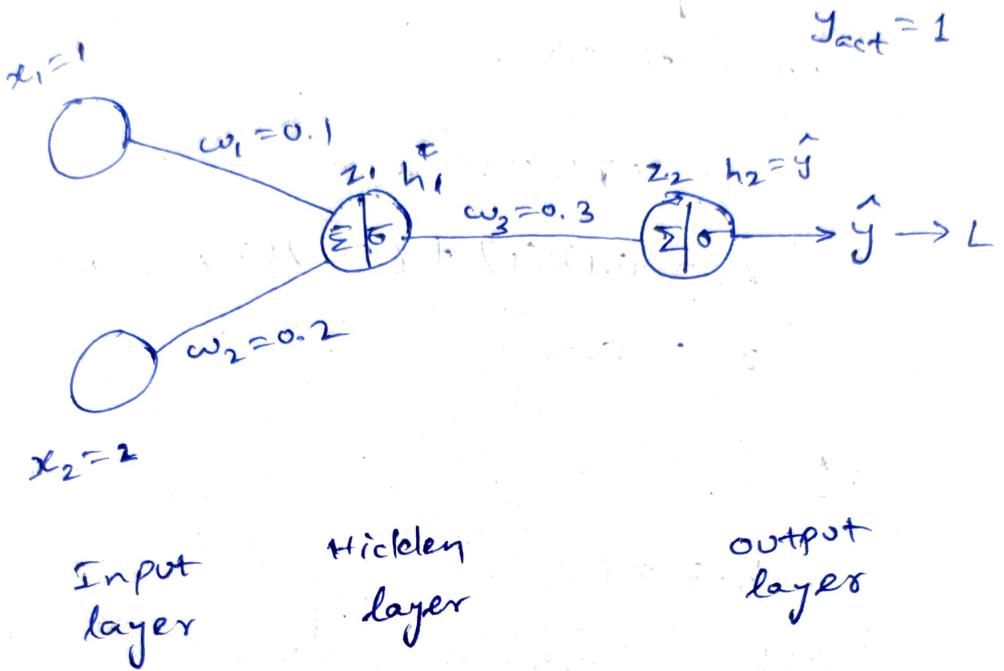


Task -1

Back Propagation



Given:

$$x_1 = 1$$

$$x_2 = 2$$

$$y_{act} = 1$$

$$w_1 = 0.1$$

$$w_2 = 0.2$$

$$w_3 = 0.3$$

$$\text{Learning Rate} = \eta = 0.1$$

Calculate. Sigmoid & Activation MSE loss

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\text{MSE loss}(L) = \frac{1}{2} (y_{act} - \hat{y})^2$$

Q.1. Forward Pass and Predicted Output.

Hidden layer :

pre-Activation :

$$z_1 = w_1 x_1 + w_2 x_2$$

$$z_1 = (0.1)(1) + (0.2)(2) = 0.5$$

$$\boxed{z_1 = 0.5}$$

Activation :

$$h_1 = \sigma(z_1)$$

$$h_1 = \sigma(0.5) = \frac{1}{1 + e^{-0.5}}$$

$$h_1 = \frac{1}{1 + 0.60653} = \frac{1}{1.60653}$$

$$h_1 = 0.62246$$

$$\boxed{h_1 = 0.62246}$$

output layer :

pre-Activation:

$$z_2 = h_1 \cdot w_3$$

$$z_2 = (0.62246) \times (0.3)$$

$$z_2 = 0.186738$$

$$\boxed{z_2 = 0.186738}$$

Activation: (Predicted output $\hat{y}$)

$$h_2 = \sigma(z_2) \quad (\because \hat{y} = h_2)$$

$$h_2 = \sigma(0.186738)$$

$$= \frac{1}{1 + e^{-0.186738}}$$

$$h_2 = \frac{1}{1 + 0.82975} = 0.54655$$

$$\therefore \boxed{\hat{y} = h_2 = 0.54655}$$

Q.2. compute the Loss:

$$L = \frac{1}{2} (y_{\text{act}} - \hat{y})^2$$

$$= \frac{1}{2} (1 - 0.54655)^2$$

$$= 0.10281$$

$$\therefore \boxed{L = 0.10281}$$

Q.3. compute $\frac{\partial L}{\partial \omega_1}$, $\frac{\partial L}{\partial \omega_2}$, $\frac{\partial L}{\partial \omega_3}$

[Backward Pass]

$$\textcircled{1} \frac{\partial L}{\partial \omega_1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_2} \times \frac{\partial z_2}{\partial h_1} \times \frac{\partial h_1}{\partial z_1} \times \frac{\partial z_1}{\partial \omega_1}$$

$$\textcircled{2} \frac{\partial L}{\partial \omega_2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_2} \times \frac{\partial z_2}{\partial h_1} \times \frac{\partial h_1}{\partial z_1} \times \frac{\partial z_1}{\partial \omega_2}$$

$$\textcircled{3} \frac{\partial L}{\partial \omega_3} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_2} \times \frac{\partial z_2}{\partial \omega_3}$$

⇒ Output layer Derivatives
derivative of Loss wrt $\hat{y}$:

$$\frac{\partial L}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}} \left(\frac{1}{2} (y_{act} - \hat{y})^2 \right)$$

$$= (\hat{y} - y_{act})$$

$$= 0.54655 - 1$$

$$= -0.45345$$

$$\boxed{\frac{\partial L}{\partial \hat{y}} = -0.45345}$$

derivative of sigmoid at output :

$$\frac{\partial \hat{y}}{\partial z_2} = \hat{y} (1 - \hat{y})$$

$$= 0.54655 (1 - 0.54655)$$

$$= 0.24783$$

$$\boxed{\frac{\partial \hat{y}}{\partial z_2} = 0.24783}$$

derivative of loss wrt z_2

$$\frac{\partial L}{\partial z_2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_2}$$

$$= (-0.45345) \times (0.24783)$$

$$= -0.11229$$

① Gradient wrt w_3 : $[w_3 \rightarrow z_2 \rightarrow \hat{y} \rightarrow L]$

$$\text{since, } z_2 = h_1 \cdot w_3$$

$$\frac{\partial z_2}{\partial w_3} = h_1$$

$$\text{then, } \frac{\partial L}{\partial w_3} = \left(\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_2} \right) \times \frac{\partial z_2}{\partial w_3}$$

$$\frac{\partial L}{\partial w_3} = \left(\frac{\partial L}{\partial z_2} \right) \times \frac{\partial z_2}{\partial w_3}$$

$$= (-0.11229) \times (0.62246)$$

$$= -0.06990$$

$$\therefore \boxed{\frac{\partial L}{\partial w_3} = -0.06990}$$

② gradient w.r.t w_1 :

$$[w_1 \rightarrow z_1 \rightarrow h_1 \rightarrow z_2 \rightarrow \hat{y} \rightarrow L]$$

$$\frac{\partial z_2}{\partial h_1} = \frac{\partial}{\partial h_1} (w_3 \cdot h_1) = w_3$$

$$\frac{\partial z_2}{\partial h_1} = 0.3$$

$$\begin{aligned} \frac{\partial h_1}{\partial z_1} &= h_1(1-h_1) \\ &= (0.62246)(1-0.62246) \\ &= 0.23500 \end{aligned}$$

$$\frac{\partial h_1}{\partial z_1} = 0.23500$$

$$\frac{\partial z_1}{\partial w_1} = \frac{\partial}{\partial w_1} (x_1 \cdot w_1) = x_1$$

$$\frac{\partial z_1}{\partial w_1} = 1$$

then,

$$\frac{\partial L}{\partial w_1} = (-0.11229) \times (0.3) \times (0.23500) \times (1)$$

$$= -0.00792$$

$$\therefore \boxed{\frac{\partial L}{\partial w_1} = -0.00792}$$

⑧ Gradient w.r.t w_2 :

$$[w_2 \rightarrow z_1 \rightarrow h_1 \rightarrow z_2 \rightarrow \hat{y} \rightarrow L]$$

similar part of calculation as w_1

but final term is $x_2 = 2$

$$\frac{\partial z_1}{\partial w_2} = \frac{\partial}{\partial w_2} (x_2 \cdot w_2)$$

$$\frac{\partial z_1}{\partial w_2} = x_2$$

$$\Rightarrow \frac{\partial z_1}{\partial w_2} = 2$$

then, $\frac{\partial L}{\partial \omega_2} = (-0.11229) \times (0.3) \times (0.23500) \times (2)$

$$= -0.01584$$

$$\therefore \boxed{\frac{\partial L}{\partial \omega_2} = -0.01584}$$

Conclusion of Question-3

$$\frac{\partial L}{\partial \omega_1} = -0.00792$$

$$\frac{\partial L}{\partial \omega_2} = -0.01584$$

$$\frac{\partial L}{\partial \omega_3} = -0.06990$$

Q.4. update all weights using
Gradient Descent

Learning Rate (η) = 0.1

Gradient Descent update Rule

$$w_{\text{new}} = w_{\text{old}} - \eta \cdot \frac{\partial L}{\partial w}$$

① update w_1 :

$$w_{1,\text{new}} = w_{1,\text{old}} - \eta \cdot \frac{\partial L}{\partial w_1}$$

$$= (0.1) - (0.1) \times (-0.00792)$$

$$= (0.1) + (0.000792)$$

$$= 0.100792$$

$$\therefore w_1^{\text{new}} = 0.100792$$

② update ω_2 :

$$\omega_2^{\text{new}} = \omega_2^{\text{old}} - \eta \cdot \frac{\partial L}{\partial \omega_2}$$

$$= (0.2) - (0.1) \times (-0.01584)$$

$$= 0.2 + 0.001584$$

$$= 0.201584$$

$$\therefore \boxed{\omega_2^{\text{new}} = 0.201584}$$

③ update ω_3 :

$$\omega_3^{\text{new}} = \omega_3^{\text{old}} - \eta \cdot \frac{\partial L}{\partial \omega_3}$$

$$= (0.3) - (0.1) \times (-0.06990)$$

$$= 0.3 + 0.006990$$

$$= 0.30699$$

$$\therefore \boxed{\omega_3^{\text{new}} = 0.30699}$$